Software Engineering Lab/ 3th Grade

[Second Semester] and [2026]

[Lab 3] [17/2/ 2026] [8:30 + 10:30] am] Instructor Information

Instructor
Lec. Najwan Abed
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✓ Using Boolean logic ✓
Branching code
✓ Looping code

Flow Control

Boolean Logic The Foundation

Boolean logic forms the basis of all decision-making in code. The bool type holds only two values: true or false.
Named after mathematician George Boole, this system enables programs to make intelligent decisions based on conditions.

Boolean Comparison Operators

OPERATOR	CATEGORY	EXAMPLE EXPRESSION	RESULT
==	Binary	var1 = var2 == var3;	var1 is assigned the value true if var2 is equal to var3, or false otherwise.
!=	Binary	var1 = var2 != var3;	var1 is assigned the value true if var2 is not equal to var3, or false otherwise.
<	Binary	var1 = var2 < var3;	var1 is assigned the value true if var2 is less than var3, or false otherwise.
>	Binary	var1 = var2 > var3;	var1 is assigned the value true if var2 is greater than var3, or false otherwise.
<=	Binary	var1 = var2 <= var3;	var1 is assigned the value true if var2 is less than or equal to var3, or false otherwise.
>=	Binary	var1 = var2 >= var3;	var1 is assigned the value true if var2 is greater than or equal to var3, or false otherwise.

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Conditional Boolean Operators

OPERATOR	CATEGORY	EXAMPLE EXPRESSION	RESULT
&&	Binary	var1 = var2 && var3;	var1 is assigned the value true if var2 and var3 are both true, or false otherwise. (Logical AND)
||	Binary	var1 = var2 || var3;	var1 is assigned the value true if either var2 or var3 (or both) is true, or false otherwise. (Logical OR)

Performance Advantage

Conditional operators improve performance by skipping unnecessary evaluations

Use && and || instead of & and | for better efficiency

1- Highest Priority: Prefix operators: ++, --, unary +, -, !, ~

2- Arithmetic Operations: Multiplication, division, modulus, then addition and subtraction 3- Comparison Operators: Relational operators: <, >, <=, >=, ==, !=

4- Logical Operations: Bitwise, then conditional: &, ^, |, &&, ||

5-Assignment Operators: All assignment operations including compound assignments

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Operator Precedence

PRECEDENCE	OPERATORS
Highest	++, -- (used as prefixes); (), +, - (unary), !, ~
	*, /, %
	+, -
	<<, >>
	<, >, <=, >=
	==, !=
	&
	^
	|
	&&
	||
	=, *=, /=, %=, +=, -=, <<=, >>=, &=, ^=, |=
Lowest	++, -- (used as suffixes)

Boolean Logic & Clarity

Use Parentheses for Clarity

Compiler knows operator order, but humans forget. Make expressions explicit: var1 = (var2 <= 4) && (var2 >= 2);

Parentheses solve ambiguity and allow easy reordering

BRANCHING

Three Branching Techniques

Ternary Operator

Simplest comparison for quick conditional assignments

if Statement

Versatile decision-making for conditional execution

switch Statement

Multiple value testing in single structure

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1- Ternary Operator

Syntax

```
<test> ? <resultIfTrue> : <resultIfFalse>
```

Works on three operands, returns one of two values

Example

```
string resultString = (myInteger < 10)
  ? "Less than 10"
  : "Greater than or equal to 10";
```

Evaluates test, assigns appropriate string based on result

2-The if Statement

1- Simple if Execute code only if test is true

2- if-else Execute different code for true vs. false 3- Code blocks Use braces for multiple lines in each branch

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if Statement Example

Simple if:

```
if (<test>)
    <code executed if <test> is true>;
```

If-else Block

```
if (<test>)
{
  <code executed if <test> is true>;
}
else
{
  <code executed if <test> is false>;
}
```

if-else

```
if (<test>)
    <code executed if <test> is true>;
else
    <code executed if <test> is false>;
```

Nested if approach

```
if (var1 == 1)
{
  // Do something.
}
else
    {
        if (var1 == 2)
          {
            // Do something else.
          }
          else
              {
                if (var1 == 3 || var1 == 4)
                  {
                    // Do something else.
                  }
                  else
                      {
                        // Do something else.
                      }
              }
    }
```


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Basic Structure of switch statement

```
switch (<testVar>)
        {
                case <comparison_val_1>:
                                        < code to execute if <testVar> == <comparison_val_1> >
                                        break;
                case <comparison_val_2>:
                                        < code to execute if <testVar> == <comparison_val_2> >
                                        break;
                …
                …
                case <comparison_val_n>:
                                        < code to execute if <testVar> == <comparison_val_n> >
                                        break;
                default:
                        < code to execute if <testVar> != <comparison_vals> >
                         break;
        }
```